AKSHITH KATKURI

Fullstack AI Engineer

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SUMMARY

Fullstack AI and systems engineer with 7+ years of programming and 4 years of professional experience building and launching production platforms from 0 to 1. Former technical co-founder of GaurData: built a governed semantic layer and agent runtime in TypeScript (TanStack Start/AI), vectorized Rust query execution with rig.rs, per-tenant DuckDB analytical isolation over gRPC, and multi-protocol ingestion under 10 minutes. At Factly, served as Project Lead for Gopie (multi-agent nl2sql in FastAPI/LangGraph cutting multi-OLAP latency from 5s to <500ms across DuckDB and ClickHouse), built Meterus (high-availability telemetry in Rust ingesting millions of events/sec with 99.99% uptime), and built Tagore AI (Go streaming research platform indexing 2M+ documents). Proven builder across Rust, Go, Python, TypeScript, modern web, databases, and deterministic production AI agents.

WORK EXPERIENCE

GaurData, Co-Founder

Feb 2026 – Aug 2026

Hyderabad, India · gaur.run · Co-Founder & Sole Technical Lead

- Handled tech and product end to end as solo technical founder, taking customer data from raw upload to an interactive queryable application in under 10 minutes at Gaur (gaur.run).
- Built the platform surface solo from scratch in TypeScript on TanStack Start and TanStack AI with server functions, SSR, SSE streaming, and session authentication.
- Architected a unified Rust backend integrating vectorized query execution with rig.rs agent runtimes, isolating analytical queries into per-tenant DuckDB processes over async gRPC (tonic).
- Built a three-protocol consumption layer: structured REST query API, OpenAI-compatible chat endpoint streaming custom delta.gaur_event payloads, and an HTTP streamable MCP server for AI agents.
- Engineered multi-source connectors (S3, Cloudflare R2, Postgres, MySQL, SQLite, Stripe, Webhooks) with automated schema inference and DuckDB indexing for full and incremental syncs under 10 minutes.
- Implemented declarative semantic contracts enforcing single-fact rules, fan-out/chasm prevention, row-level security (RLS), and cryptographic audit logging.

Factly Research & Media

Sep 2022 – Jan 2026

Hyderabad, India

Senior Backend Developer & Project Lead May 2024 – Jan 2026

- Built Gopie (gopie.ai) end to end: engineered high-concurrency backend services in Go using hexagonal architecture and sqlc, and built the multi-agent nl2sql runtime in FastAPI with LangGraph.
- Built supervisor agent routing between single and multi-dataset sub-graphs with semantic schema retrieval in Qdrant, query decomposition, and parallel execution.
- Optimized multi-OLAP query routing across DuckDB and ClickHouse, slashing analytical query latency from 5s down to under 500ms.
- Integrated E2B secure microVM sandboxes for safe execution of dynamic Python transformations and built automated Vega-Lite visualization generation.
- Built and operated Meterus, a high-availability telemetry and metering engine in Rust benchmarked to process millions of events per second via Redpanda into ClickHouse with 99.99% production uptime.
- Built low-latency gRPC quota validation with in-memory caching and Redis fallbacks, evaluating customer quotas and API key entitlements in under 1 millisecond.
- Instrumented query pipelines with LangSmith for prompt versioning, automated evaluation benchmarks, and systematic accuracy regression tracking.
- Mentored junior engineers and interns through regular pairing sessions and code reviews, establishing hexagonal architecture and testing conventions across Go, Rust, and TypeScript codebases.

Back End Developer Feb 2023 – Apr 2024

- Built the Go real-time streaming backend and LangGraph multi-step research agent behind Tagore AI (tagore.ai), delivering cited, zero-hallucination answers across 2,000,000+ documents and 500,000+ tables.
- Implemented SSE token streaming multiplexed with citation payloads, plus hybrid vector retrieval combining Qdrant dense embeddings and pgvector sparse search.
- Architected and deployed a centralized chat history microservice in Go and MongoDB to persist conversational state, session context, and citation metadata across internal AI products.
- Integrated Portkey and LiteLLM as an enterprise AI gateway layer for intelligent model routing, rate limiting, latency tracking, and automated multi-provider fallbacks.

- Maintained and implemented performance optimizations for Kavach, Factly's internal authentication and authorization system built on the Ory stack (Kratos, Keto, and Oathkeeper).
- Scaled ruspie in Rust to query 25,000+ public datasets straight from Google Cloud Storage with an nl2sql endpoint, and converted public catalogs from CSV to Apache Parquet.

Back End Developer Intern Sep 2022 – Jan 2023

- Built ruspie from scratch in Rust on top of Apache Arrow and DataFusion to query CSV and Parquet files directly from cloud storage with zero database pre-loading.
- Shipped ruspie as the core query engine powering dataful.in (public dataset marketplace) and open-sourced it at github.com/factly/ruspie.
- Contributed upstream performance enhancements and cloud storage adapter fixes to roapi (roapi/roapi).

PROJECTS

- **rocky** (github.com/elliot14A/rocky) – author. Opinionated NixOS fleet deployment CLI tool in Rust managing VPS instances across Hetzner and OCI with automated Disko partitioning, Tailscale mesh networking, Agenix secret scoping, and zero-downtime rollbacks via deploy-rs.
- **elliot14a.work** (github.com/elliot14A/portfolio) – author. Type-safe personal portfolio and streaming AI agent built with Effect 4.x, Preact SSR, and Vanilla Extract zero-runtime CSS, deployed to a single Cloudflare Worker with D1 sliding-window rate limiting.
- **fincher** (github.com/elliot14A/fincher) – author. Autonomous operations engine for media post-production release pipelines with asynchronous workflow graphs in Go, live SSE trace triage, read-only AI agents querying ClickHouse/SQLite over MCP, and deterministic policy verification.
- **cerberust** (github.com/elliot14A/cerberust) – author. High-reliability Rust authentication server built on Axum, Diesel, and PostgreSQL with JWT token rotation, hierarchical resource trees, and Role-Based Access Control (RBAC).
- **ruspie** (github.com/factly/ruspie) – author. In-place columnar query engine in Rust on Apache Arrow and DataFusion serving 25,000+ datasets over SQL, GraphQL, and REST. Open-sourced and in production at Factly.
- **minitraycer** (github.com/elliot14A/minitraycer) – author. TypeScript CLI and MCP task-planning server that gates autonomous coding agents behind human-approved verification plans.

SKILLS

Core Programming Languages: TypeScript & JavaScript (Node.js, modern ECMAScript, React, TanStack Start, Bun), Rust (Tokio, async gRPC/Tonic, Apache Arrow, DataFusion, rig.rs, Axum, Diesel), Go / Golang (goroutines, channels, microservices, gRPC, sqlc, hexagonal architecture, Echo), Python (Python 3.11+, FastAPI, asyncio, LangGraph, LangSmith, pytest), SQL (PostgreSQL, schema design, query tuning, indexing, migrations, RLS), Bash & Systems Scripting

AI Platforms & Agent Systems: Agent Frameworks: LangGraph, rig.rs, Google ADK, Model Context Protocol (MCP) servers, Evaluation & Gateways: LangSmith evals, Portkey, LiteLLM, prompt versioning, systematic regression benchmarks, Vector & Retrieval: pgvector, Qdrant dense/sparse embeddings, semantic schema search, Execution & Streaming: E2B microVM secure code sandboxes, Server-Sent Events (SSE), token multiplexing

Backend, Systems & Infrastructure: Architecture & Frameworks: Effect-TS (4.x), FastAPI, Tonic (gRPC), Echo, Axum, Node.js, Protobuf, Databases & Engines: PostgreSQL, DuckDB, ClickHouse, Redis, MongoDB, Apache Arrow, Parquet, Event Streaming & Telemetry: Redpanda (Kafka API), CloudEvents (v1.0), distributed message partitions, Identity & Governance: Ory Stack (Kratos, Keto, Oathkeeper), OAuth, API key scoping, RBAC, tenant isolation, DevOps & Cloud: Cloudflare Workers & D1, Docker, Nix / NixOS (Rocky tool), Hetzner, AWS S3 / Cloudflare R2, CI/CD

EDUCATION

Bachelor of Technology (B.Tech), Electronics and Computer Engineering Jun 2019 – Jun 2023

Sreenidhi Institute of Science and Technology, Hyderabad, India (affiliated to JNTU Hyderabad)

Languages: English (C1) · Hindi · Telugu · German (A1, in progress)